

Mash pH control

From German brewing and more

This is the 3rd article in a series of articles intended to educate the interested brewer about pH in brewing. Part 1, An Overview of pH, showed the basic principles behind pH and how it can be measured, part 2, How pH affects brewing, illustrated the effects pH has on various brewing processes and which pH ranges are considered optimal. This article will be the most practical. It shows how malt and water settle at a pH and how this pH can be affected through changes in water composition, malt bill and mash additions. For the interested brewer, it also goes behind the scenes and explains the chemical processes that are at work. A number of water treatment and brewing water building options are explained.

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Water, malt and mash pH

Water and malt are both pH buffers, that means they have their own pH and a desire to resist pH changes. When they are brought together at dough-in, they will settle at a pH which will be the mash pH. Which pH they'll settle at depends on how strong either of the components (water and malt) pull pH to their respective sides. In this "tug of war" malt is the acidic one, it wants to lower pH, and water tends to be the alkaline component, it wants to raise pH.



As we have seen in Enzymatic Activity the mash pH range that works for brewing is fairly wide. 5.0 – 6.0 will work with most enzymatically strong malts and 5.3-5.5 is considered optimal. This wide range of possible mash pH values is the reason why most brewers don't have to worry about mash pH and water chemistry at the beginning of their home brewing career. As we will see later, average water and grist compositions generally found in (home) brewing are likely to result in a mash pH in that range. Only at the extremes will brewers experience problems in mash conversion and off flavors that arise from incorrect mash pH levels which spark the brewer's interest in that rather technical topic.

But even if you don't experience off flavors or mash problems from incorrect mash pH, there is likely a point in your brewing career where the beers are good and very enjoyable but you want to take them to excellent. Controlling your mash pH and moving it into a more optimal range has the potential to do that for you. The effects that pH has in brewing were elaborated in How pH affects brewing and they start with a proper mash pH.

A correct mash pH is all about matching the right water with the right grist and possibly the addition of some acid. Since we brewers generally start with a style of beer or recipe in mind, the grist composition is known and we want to know what kind of water modifications are needed to get the desired mash pH. In a way this is like balancing a scale (Figure 1). On the left is the malt and on the right is the water. But we are not interested in their actual weight. Instead we want to know how acidic the malt is and how alkaline is the water.



Figure 1 - Balancing malt and water is, in its simplest form, like balancing a scale

To determine how acidic malt is, I ran experiments where I tested the distilled water mash pH of some base malts and the acidity of various specialty malts. And what I found was that darker malts are generally more acidic than lighter colored malts. While there were a few exceptions to that rule, which we will examine later, we will work with that for now: The darker malts in the grain bill, the more acidic the grist is.

That acidity of the malt is counteracted by the alkalinity of the water. Water alkalinity measures the water's ability to neutralize acids and thus resist a change in pH. In fact, it is measured by adding a strong acid to a sample of water until it reaches a pH of 4.3 [1]. The amount of acid needed determines how alkaline the water is. The more acid that is needed, the more alkaline the water is.

When grist and water are mixed the alkalinity of the water will neutralize some of the malt's acidity. Neutralization happens when an acid and a base are brought together. The result is a salt and water and a pH that lies between the pH of the acid and the pH of the base. The acidity of the grist is best thought of as its mash pH with distilled water. Alkalinity present in the water or from the addition of alkaline salts pulls that pH higher (more alkaline) while the addition of acid, calcium or magnesium lowers that pH (more acidic).

If, for example, the grist has low acidity because it contains only lightly colored malts and the water has a lot of alkalinity, the mash pH will be higher than desired. If the opposite is true, lots of malt acidity from dark grains and low alkalinity water, the mash pH will be lower than desired. If malt acidity and water alkalinity are balanced, the pH will be in the desired range.

Mash as a pH buffer



Grist, water, acid and salt interactions within the mash are best thought of as the addition of acids or bases to a strong buffer which is largely formed by phosphates, proteins and amino acids present in the malt. Dealing with a buffered system means that the amount of acid or base needed to change pH depends largely on the amount of buffer (in this case weight of the grist) and the desired pH change. The amount of acid it takes to lower or raise the mash pH of 1 kg of grist by 0.1 pH units is about 3-5 mEq. A simple Model for pH Buffers shows how we can visualize pH buffers as differently shaped vertical tubes.

mEq (milliequivalent) or Eq (1 mEq = 0.001 Eq) is a unit commonly used to express the amount of acid or base. Rather than giving the specific amount of acid or base needed it specifies the number of hydrogen ions that are added (acid) or absorbed (base). To convert Eq to an actual amount of acid/base its molar weight, number of hydrogen ions donated/absorbed and its concentration needs to be known. Lactic acid for example has a molar weight of 90 g/mol. Above a pH of 5 almost all its molecules donate 1 hydrogen ion. This means 90g 100% lactic acid contribute about 1 Eq acid. The lactic acid we commonly use has a concentration of 88% by weight and 1 Eq of that weighs 102 g. 1 mEq weighs 0.102 g.

Calcium carbonate (chalk), on the other hand, has a molar weight of 100 g/mol but can absorb 2 hydrogen ions. To get 1 Eq worth of acid neutralizing power 50g chalk are needed. In brewing that neutralizing power will be lower due to the acidifying effect of the calcium when it reacts with the malt's phosphates. But more about this later.

Table 1 on the right shows the amounts of a few acids and bases that provide 1 mEq of neutralizing power.

The buffer capacity of the malt, which affects the amount of acid or base needed to change mash pH, does to some extent depend on the malting process as well as the mashing process. This may lead to differences how the mash responds to pH correction efforts and makes exact mash pH predictions difficult.

1 mEq worth of an acid is		
0.085	ml	88% lactic acid
3	g	acidulated malt (Sauermalz)
0.98	ml	10% phosphoric acid
0.081	ml	37% Hydrochloric acid
1 mEq worth of a base is		
0.050	g	chalk (calcium carbonate)
0.084	g	baking soda (sodium bicarbonate)
0.037	g	slaked lime (calcium hydroxide)
0.040	g	caustic soda (sodium hydroxide)
0.500	l	water with 100 ppm as CaCO3 alkalinity
0.132	gal	water with 100 ppm as CaCO3 alkalinity

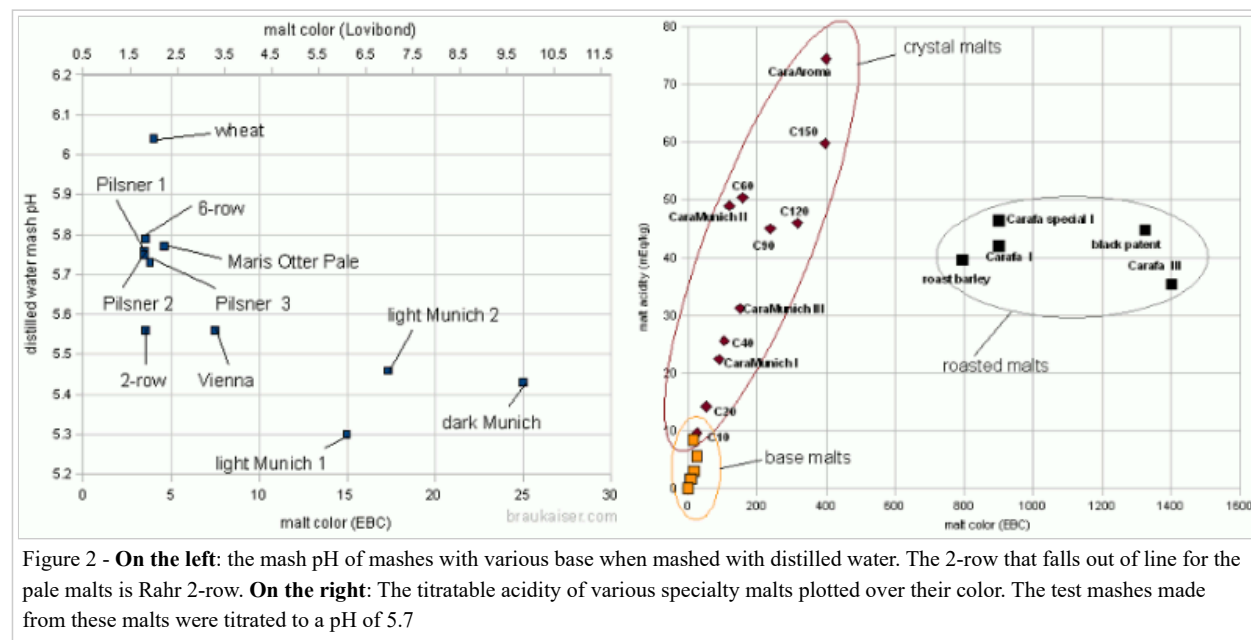
Table 1 - Examples for 1 mEq worth of acid and base. A pH of ~5.5 is assumed



Factors affecting mash pH

The following is a more detailed discussion of what affects mash pH in brewing.

Grist



Very lightly kilned malts have a natural pH between 5.7 and 5.8. Wheat malts may even be higher than that. Once malts are kilned more intensely to produce darker base malts, the distilled water pH drops due to the formation of acidic melanoidins. This is shown in Figure 2 which plots the color and distilled water mash pH for a number of different base malt samples. There is a loose correlation between malt color and mash pH: "The darker the malt the lower the mash pH". Because of this loose correlation the prediction of the malts mash pH based on its color is difficult in particular for darker malts.

Specialty malts like crystal and roasted malts are kilned to even darker colors than base malts. While they may constitute a much smaller portion of the grist they tend contribute a large portion of the grist's color and with it acidity. The latter lowers the grist pH below the pH of the base malts used.

The acidity of various specialty malts is plotted over their color in Figure 2. Crystal type malts adhere fairly strongly to the "the darker the more acidic" rule. Roasted malts, on the other hand, showed a fairly constant acidity regardless of their color. This is assumed to be the result of differences in the production process for crystal and roasted malts. The color of crystal malts is created while the malt is still wet which allows for the formation of more acidic compounds while the color of roasted malts is created after they have already been dried. Though that creates a stronger color it creates less acidity^[2]

For us brewers this means that grists containing large amounts of roasted malts are likely less acidic than grists containing large amounts of dark crystal malts even though the beers made with roasted malts might be darker in color.

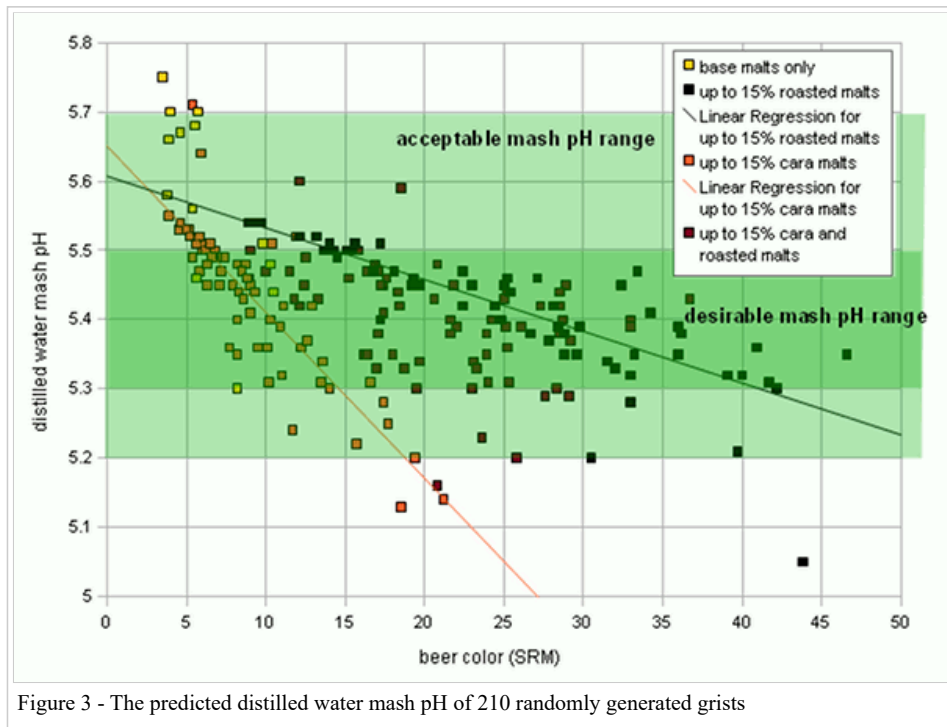


Figure 3 - The predicted distilled water mash pH of 210 randomly generated grists

To get a better understanding what crystal and roasted malts do to the acidity of the grist I randomly generated 210 different grist compositions using the malts I analyzed and the following constraints:

- 15 recipes mixed only base malts
- 45 recipes mixed base malts with up to 15% crystal malts. 50 of these recipes used only 2-row as base malt. The other 15 used a random mix of different base malts. The percentage of crystal malts in these grists was randomly chosen between 1 and 15%. In addition to that the types and percentages of different crystal malts were also chosen randomly
- 45 recipes mixed base malts with up to 8% crystal malts and up to 8% roasted malts. 50 of these recipes used only 2 row as base malt while the other 15 used a mix of base malts. Similar to the recipes that used only crystal malts as specialty malts these recipes here had their actual crystal and roasted malt percentages determined randomly. The same applied to the types of malts that were chosen
- 45 recipes used only roasted malts. These recipes were created similar to the crystal only recipes mentioned earlier except that they only used roasted malts as base malts.

The beer color for all these beers was calculated and mash pH was estimated based on the malt data available. The mash pH was then plotted over the beer color and is shown in Figure 3. It is apparent that the majority of the beers falls into the desired mash pH range if the water has no residual alkalinity (What residual alkalinity is will be explained later in this article). If the waater has moderate residual alkalinity the mash pH will be slightly but not dramatically higher. This is the reason why most average beers can be brewed well with average water. Only at the extremes would be mash pH become a problem.

Another observation in this figure is that there is a somewhat linear relationship between beer color and mash pH. The slope of that line is steeper for beers that get all or most of their color from crystal malts and flatter for beers that get their color from roasted malts. This relationship can be used to predict grist acidity from the beer color and the percentage of color that is coming from roasted malt.

Eventually this can be used to predict mash pH and such an algorithm is described in Beer color, alkalinity and mash pH and has been implemented in the Kaiser_water_calculator.xls

(https://web.archive.org/web/20240805161308/http://braukaiser.com/downloads/Kaiser_water_calculator.xls).

Water minerals

The minerals listed here may either come from the base water, from salts added to the water before dough-in or salts added to the mash. Experiments have shown that the source of the minerals does not make much of a difference in their effect on mash pH.

What makes a difference though, is the amount of minerals that are brought into the mash. This amount depends on both mash thickness (how much water per unit of grain) and the mineral concentration in the water. In other words, water in a thin mash will be able to have larger effect on mash pH compared to the same water water in a thick mash since in the latter case there will be less water, and with it less minerals, per grain.

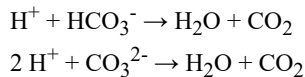
This being said, brewers should not conclude that thick mashes provide a practical means of dealing with high alkalinity water just because the alkalinity won't be able to move the mash pH as much. Any water that is not used in the mash is used during the sparge where it is able to adversely affect pH in both the grain bed and boil kettle.

This section focuses only on the water minerals which have an effect on pH. A more complete discussion of brewing water composition can be found in How to read a water report.

Bicarbonate and Alkalinity

As mentioned in the beginning, alkalinity is the water's ability to neutralize acids and the ions involved with that neutralization are bicarbonate (HCO_3^-) and carbonate (CO_3^{2-}). More info about carbonate chemistry can be found in About chalk and the carbonate system

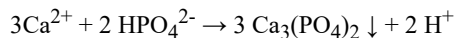
When brought in contact with the more acidic malt the bicarbonate and carbonate ions absorb hydrogen ions which results in a pH rise. The extent of this pH increase largely depends on the amount of malt, amount of water and the bicarbonate and carbonate concentration in that water. The latter is expressed by the water's alkalinity. The more bicarbonate and carbonate ions there are per unit of malt the more the pH will rise. When bicarbonate or carbonate neutralizes an hydrogen ion it forms carbon dioxide. While bicarbonate is able to neutralize one hydrogen ion carbonate can neutralize two:



Due to the poor solubility of calcium carbonate and the presence of calcium in most waters, appreciable amounts of carbonate are rarely found in brewing water. As a result the bicarbonate concentration in water can be calculated from the alkalinity of the water and vice versa. Carbonate ions will be present when chalk is added to the water or the mash. The addition of chalk, however, will be discussed later.

Calcium and Magnesium

Brewing water also contains calcium and magnesium ions. These ions are able to react with phosphates from the malt to form insoluble phosphate salts which precipitate. At mash pH values between 5 and 6 most of the phosphate is available as HPO_4^{2-} . The reaction with calcium liberates hydrogen ions which react acidic and lower the pH of the mash [3]:



Magnesium shows a similar but less effective reaction with phosphates which is why it has only half the pH lowering power of calcium.

Malt contains about 1 % of phosphate by weight^[4]. About 80% of that end up in the wort. This amount is far greater than the calcium or magnesium that is brought in with the brewing liquor which makes calcium and magnesium the limiting factor.

As early as 1914 did the German brewing scientist Windish show that water ions have an effect on the mash pH. Kolbach, another German brewing scientist, later showed that it takes about 1.75 calcium ions and twice as many magnesium ions to produce one hydrogen ion. (1.75 calcium ions equal 3.5 calcium equivalents)

My own research on this subject reported slightly lower numbers: 1.3-1.5 Calcium ions and 2.4-2.8 magnesium ions^[2]. It should be noted that Kolbachs work was primarily targeted at the cast out wort pH and not so much mash pH. There is however a close correlation between the two.

Residual Alkalinity

The acidic reaction of calcium and magnesium counteracts the alkaline reaction of the water's alkalinity, which prompted Kolbach to define the residual alkalinity as the alkalinity that remains after the calcium and magnesium reactions have been considered. Based on his work the following formula for residual alkalinity RA has been established in the brewing world [5]

$$RA = A - \frac{CH + 0.5 \cdot MH}{3.5}$$

Where:

- RA: residual alkalinity given as an equivalent unit (mEq/l, ppm as CaCO_3 , dH)
- A: alkalinity given in the same equivalent unit
- CH: calcium hardness of the water, which is the calcium ion concentration given in an equivalent unit
- MH: magnesium hardness of the water, which is the magnesium ion concentration given in an equivalent unit

The residual alkalinity is a property of the brewing water and allows brewers to estimate a water's effect on the mash pH.

If only alkalinity and general hardness (GH) are known the residual alkalinity can be estimated as:

$$RA = A - \frac{GH}{4}$$

This makes the assumption that about 30% of the water hardness comes from magnesium and the remaining 70% come from calcium, which tends to be the average split between calcium and magnesium hardness in typical waters (see Estimating Residual Alkalinity).

Water with a residual alkalinity of 0 gives about the same mash pH as distilled water while water with a RA greater than 0 yields a higher mash pH. If the water's alkalinity is low but its calcium and magnesium levels are high the residual alkalinity can be less than 0. In this case the use of that water will yield a lower mash pH.

It should be noted that Kolbach's work focused on the pH of the knock-out wort and not mash pH. While there is a close relation his finding that a residual alkalinity change of 10 dH (degrees German hardness, about 171 ppm as CaCO_3) causes a 0.3 pH change in pH applies to a 12 Plato cast out wort and not mash pH. The change in mash pH will be less and also depends on mash thickness.

Residual Alkalinity illustrated explains the concept of residual alkalinity through a few drawings of ions in the water rather than through chemical formulas alone.

Acids

We may also add acids to the water or mash to lower mash pH below the pH that is established by water and malt alone. This is in particular necessary for lighter beers where the grist itself is not acidic enough to establish the correct pH.

Acids donate hydrogen ions which neutralize the alkalinity of the water. Once all the water's alkalinity is consumed the surplus of hydrogen ions counts towards lowering the mash pH.

Other factors

The mash pH can also be affected by the chosen mash schedule. The boiling of mash in decoction mashing, for example, lowers the mash pH. I have seen a pH drop of up to 0.2 pH units during decoction mashing. This decrease of mash pH might be attributed to the enhanced precipitation of calcium and magnesium phosphates [3].

A rest between 30 and 40 C (90-110 F), known as acid rest, promotes phytase activity which releases more phosphates into the mash. While this is said to acidify the mash I have not observed a significant mash pH decrease though the use of this rest.

Strategies for affecting mash pH



When the expected or tested mash pH is outside the desired range of 5.3-5.5 the brewer may choose to change that pH. This can be done by:

- changing the grist
- changing the water
- designing water from scratch
- addition of acids or salts to the mash

Changing the grist

There was a time when brewers didn't know about water chemistry. They used the available local water and noticed that some beers turned out better than others. This led to the development of local beer styles. But these days we brewers don't want our water to dictate what beer styles we should brew. As a result changing the grist is rarely seen as a viable option.

The only case where a grist change is practical is the use of acidulated malt or Sauermalz. Acidulated malt is Pilsner malt that has been sprayed with lactic acid before it is dried again. The final lactic acid content of this malt is about 3% by weight. Each 1% of acidulated malt in the grist lowers the mash pH by ~0.1 pH units. More than 4-5% should not be used in order to prevent excessive lactic acid amounts that may be noticeable in the final beer taste.

The use of Sauermalz is an elegant way of complying with the German purity law (Reinheitsgebot) since its lactic acid has been produced by lactic acid bacteria that are naturally occurring on the surface of the malt. Because of that the mash pH can be corrected and the beer can still be brewed with just water, malt, hops and yeast. Some German brewers believe that if this loophole would not have

been found the German purity law would have needed to be amended since the benefits to beer quality attained from being able to acidify the mash are substantial.

More info about the use of lactic acid can be found later in Adding acids

Changing the water

Before the water can be changed we need to know more about the mineral content of the water. How to read a water report explains in detail the various minerals found in drinking water. It also explains what to look for in a water report.

When the water is changed to correct the mash pH it's residual alkalinity is changed to match the needs of the grist composition used in the recipe. The desired residual alkalinity range can be estimated from the beer's color. The darker the beer the more residual alkalinity will be needed to counteract the acidity of the grist.

The residual alkalinity's effect on pH also depends on mash thickness. The thinner the mash the more pronounced the effect of the water's alkalinity will be. At a mash thickness of 2 l/kg (0.95 qt/lb) it takes a residual alkalinity change of about 130 ppm as CaCO_3 (2.6 mEq/l) to change the mash pH by 0.1 pH units while it takes a residual alkalinity change of just 75 ppm as CaCO_3 (1.5 mEq/l) to achieve the same pH change at a mash thickness of 4 l/kg (1.9 qt/lb)^[2].

Raising residual alkalinity (raises mash pH)

If the water's residual alkalinity is too low the addition of alkaline salts like calcium carbonate (chalk, CaCO_3) or sodium bicarbonate (baking soda, NaHCO_3) or even strong bases like calcium hydroxide (Ca(OH)_2) raises its residual alkalinity. If calcium carbonate is used it will not dissolve in the water unless CO_2 is added. Though there are ways to dissolve calcium carbonate many brewers simply add it to the brewing water without dissolving it. In experiments I found that dissolved chalk is not only twice as effective in raising the water's residual alkalinity, undissolved chalk is also not able to raise the mash pH by more than 0.2 pH units. In other words the addition of more than 500 mg/l undissolved calcium carbonate, which is equivalent to a residual alkalinity of about 200 ppm as CaCO_3 , had little or no effect on mash pH^[2]. As shown in Beer color, alkalinity and mash pH water with a residual alkalinity of more than 200 ppm as CaCO_3 is rarely needed, even for the darkest beers. A.J. deLange, an expert in everything brewing water, believes that while the mash pH is low enough to dissolve chalk, the actual rate of dissolving chalk at this pH happens slowly which reduces its effectiveness as a means of raising mash pH^[6]. Despite its shortcomings it still remains the most popular salt addition for raising mash pH. In addition to carbonate chalk also adds calcium to the mash which counteracts some of the pH raising power of the carbonate.

Baking soda, however, is more soluble than chalk and does not show the somewhat unpredictable mash pH behavior that comes with chalk. The drawbacks of adding baking soda is the increase of the water's sodium content and the lack of calcium which has a number of positive effects on beer quality.

Another substance that can be used to increase the alkalinity of the brewing water and thus raise the mash pH is calcium hydroxide (pickling lime, slaked lime, Ca(OH)_2). It dissolves in water more readily than chalk and doesn't show the limits that undissolved chalk has while it also adds calcium to the mash. The only drawback is that it is a caustic substance and needs to be handled with care. It is best added to the mash after dough-in.

Lowering residual alkalinity (lowers mash pH)

In most cases the residual alkalinity of the water is too high for the desired beer color which causes the mash pH to be too high as well. In these cases the water's residual alkalinity needs to be reduced. For that we brewers have a number of options:

- dilution
- addition of calcium
- addition of acids to neutralize all or part of the bicarbonates
- alkalinity removal through boiling or slaked lime additions.

Dilution

The idea of dilution is simple: reduce the bicarbonate concentration, and with it the alkalinity, by mixing the water with water that contains only little bicarbonate. The water generally used for dilution is distilled or very low mineral water like reverse osmosis water. When using such water for dilution the residual alkalinity of the diluted water is

$$RA_{\text{diluted water}} = RA_{\text{base water}} \cdot \frac{r}{100}$$

Where:

- $RA_{\text{diluted water}}$: the resulting residual alkalinity
- $RA_{\text{base water}}$: the starting residual alkalinity
- r : the percentage of base water used.

Since dilution with low mineral water lowers the concentration of all minerals it may be necessary to supplement the resulting water with calcium ions to get their concentration back into the desired range of 50-150 ppm. This should be done with calcium chloride or calcium sulfate (gypsum) in order to avoid adding bicarbonate which would negate the alkalinity lowering effect of dilution. The added benefit of supplementing calcium is an additional residual alkalinity reduction through the acidic reaction between calcium and malt phosphates.

Adding Calcium

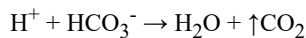
As already discussed in [[#Calcium and Magnesium]] calcium and magnesium salts are able to lower mash pH through a reaction with phosphates brought in by the malt. This is of particular interest for beer styles that benefit from water with high permanent hardness. In permanently hard water the anions (negatively charged ions) balancing the calcium and magnesium cations are chloride and sulfate which have no effect on the mash pH. This is not true for temporary hard waters where the balancing anion is bicarbonate.

Examples for these styles of beer are English Ales and Dortmunder Export. A mash water calcium content of 150 mg/l and little or no alkalinity can yield a mash pH drop of 0.1 – 0.2 pH units depending on mash thickness.

Due to its lower effectiveness with respect to changing the mash pH and its lower desired concentration in brewing water, magnesium salts are generally not used to affect mash pH in any meaningful way.

Adding acids

The water's bicarbonate content, and with it the alkalinity and residual alkalinity, can also be lowered or completely removed through the addition of organic or inorganic acids. The hydrogen ions released from these acids react with the bicarbonate to form carbon dioxide and water:



When acids are used to reduce the alkalinity bicarbonate is replaced with the anion provided by the acid. As a result, excessive use of acids, which may seem necessary with very alkaline waters, can lead to an excess of these ions and an adverse effect on taste. To prevent this water alkalinity may first need to be reduced with means that lower the water's mineral contents like dilution or alkalinity precipitation, for example.

While many acids can be used to accomplish this task only a few have found practical use for water treatment in brewing. Those acids are:

- **Lactic acid** is an organic acid produced by lactic acid bacteria. To brewers it is available as a 88% by weight solution or acidulated malt. Acidulated malt is pilsner malt that has been sprayed with lactic acid and dried. It contains about 3% lactic acid by weight. Lactic acid may also come from a sour mash or sour fermentation. The latter or acidulated malt is the only acid that can be used for mash and wort pH adjustment in Germany. Narziss reports that the use of lactic acid yields in a smoother beer taste compared to the use of organic acids like hydrochloric acid [7]. The anions left behind by lactic acid are lactates which can give the beer a sour twang if used in excess. In my own beers I have used as much as 0.25 g lactic acid per liter of beer which amounts to about 4% acidulated malt in the grist without adverse taste effects.
- **Hydrochloric acid (a.k.a muriatic acid)** is a strong inorganic acid which replaces the bicarbonate with chloride and, when used in excess, can give the beer a salty taste. The muriatic acid found in hardware and pool supply stores is not necessarily food grade and should be avoided. If food grade hydrochloric acid is available it can be used for water treatment but great care should be taken when handling it. Unlike lactic acid, hydrochloric acid is a strongly caustic acid that readily reacts with almost anything it comes in contact with, including your skin and eyes.
- **Sulfuric acid:** This strong inorganic acid replaces bicarbonate with sulfate and would therefore be a good choice for hoppiest beer styles. It is however even more aggressive and dangerous to handle than hydrochloric acid which is why it is rarely used by home brewers.
- "Phosphoric acid" is a weak inorganic acid that is much safer to handle than hydrochloric or sulfuric acid and is widely used in soft drinks. It replaces bicarbonate with phosphate and therefore increases the phosphate content of the mash. The amount of phosphate added, however, is small compared to the phosphate added by the malt and therefore the use of phosphoric acid should not lead to the precipitation of additional calcium and magnesium.

to lower the residual alkalinity by 100 ppm as CaCO ₃ it takes						
	88% lactic acid HC ₃ H ₅ O ₃ ml	acidulated malt HC ₃ H ₅ O ₃ g	37% hydrochloric acid HCl ml	18M (95-98%) sulfuric acid H ₂ SO ₄ ml	85% phosphoric acid H ₃ PO ₄ ml	10% phosphoric acid H ₃ PO ₄ ml
in 1 l	0.17	6.0	0.16	0.06	0.14	1.98
in 1 gal	0.64	22.7	0.62	0.21	0.52	7.48
in 1 bbl	20.0	703.1	19.3	6.5	16.0	232.0

Table 2 – The amount of acid it takes to lower the water’s alkalinity, and with it the residual alkalinity, by 100 ppm as CaCO₃

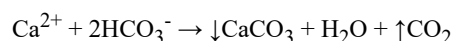
If more acid than what is needed to neutralize all alkalinity is added, the alkalinity of the water becomes negative. The acid that was not used for neutralizing alkalinity will contribute to lowering the mash pH once the malt is added to the water.

In order to get the mash pH within the desired 5.3-5.6 range when brewing very pale beers, Pilsners for example, it is generally necessary to use an acid regardless how low the residual alkalinity of the brewing water is. This is because even calcium contents on the high end (150 ppm) and an alkalinity of 0 yields a residual alkalinity of only -70 ppm as CaCO₃. This residual alkalinity can only lower the mash pH by ~0.2 units when a 4 l/kg (2 qt/lb) mash thickness is used and even less for thicker mashes. With a distilled water mash pH for pilsner malt that tends to be between 5.7 and 5.8 the resulting mash pH without acid additions would only be in the 5.5 to 5.6 range. 2% Sauermalz will get it into the for lighter beers more preferable 5.3-5.4 range.

Precipitation of calcium carbonate

Since the addition of acid does not change the total amount of minerals it may not be suited for the treatment of waters that contain high concentrations of minerals. One method suited for these waters is the precipitation of calcium carbonate through either boiling or the addition of slaked lime. The calcium, and to some extent magnesium, and alkalinity removed through these methods are called temporary hardness. Temporary hardness is the calcium and magnesium ions that can be matched up with bicarbonate ions from the water. In fact, temporary hardness is called temporary because it can be removed through boiling. The remaining calcium and magnesium, which has only sulfate and chloride to pair up with is called permanent hardness.

When water is boiled CO₂ escapes. This raises the water pH and leads to the creation of more carbonate from the existing bicarbonate. In the presence of calcium the carbonate forms calcium carbonate (a.k.a chalk, CaCO₃) which is poorly soluble and creates a white precipitate which settles out. If you have water with a high temporary hardness (lots of bicarbonate calcium and magnesium) you may have already noticed this when you cook with your water.



This method is able to lower the alkalinity to about 50 - 65 ppm as CaCO₃ [8]. The amount of alkalinity that can be removed is the difference between the current alkalinity and the lowest limit (50 - 65 ppm as CaCO₃) or the calcium hardness, whichever is lower. Sufficient calcium content (i.e. calcium hardness) is needed since the precipitation of 1 ppm alkalinity as CaCO₃ also precipitates 1 ppm calcium hardness as CaCO₃ or 0.4 mg/l calcium. As a result the water may become calcium deficient.

The final calcium level can be calculated with this formula:

$$Ca_{\text{after boiling}} = Ca_{\text{before boiling}} - \frac{A_{\text{before boiling}} - 60}{2.5}$$

Where

- Ca_{after boiling} is the water calcium concentration in mg/l (or ppm) after boiling
- Ca_{before boiling} is the initial water calcium concentration in mg/l (or ppm)
- A is the initial water alkalinity in ppm as CaCO₃

If the resulting Calcium concentration is below the recommended minimum of 50 mg/l gypsum or calcium chloride should be added to boost the initial calcium content before the water is boiled. In addition to that the addition of a small amount of chalk facilitates this precipitation by providing nucleation sites for the precipitating chalk [9].

Water exposed to atmospheric CO₂ pressure can only keep about 47 mg/l calcium carbonate (chalk) in solution [10]. This amounts to an alkalinity of about 50 ppm as CaCO₃ which is less than the lowest possible alkalinity level cited above and as a result allowing the water to stand for an extended time after it has been boiled does not risk re-dissolving the precipitated chalk though its uptake of atmospheric

CO₂.

If a GH&KH test kit is available the final hardness and alkalinity achieved through boiling can be checked and used for the estimation of the mash pH (see At home water testing).

While the aforementioned process is simple, the need to boil a large amount of water represents a great deal of wasted energy. The excess CO₂ can also be removed from the water through intensive aeration but that process takes a long time. In fact, surface water tends to have low temporary hardness because due to the lower CO₂ content in air the water's CO₂ content is much lower than that possible in ground water and as a result this water cannot hold as much calcium carbonate as ground water.

A more practical approach, that is used by many breweries, is water treatment with slaked lime. The slaked lime is able to absorb the water's CO₂ and raise the water pH to transform bicarbonate into carbonate:



How to conduct this water treatment and calculate the amount of lime needed has been described in detail in Alkalinity reduction with slaked lime and will not be discussed further.

Both boiling and slaked lime treatment will only remove bicarbonate and calcium (to some extent magnesium as well) but cannot remove sodium, chloride or sulfate. Those minerals do not affect pH but if a lower concentration is desired the only practical method the home brewer has for their removal is dilution with low mineral water or building brewing water from scratch.

Building brewing water from scratch

Using low mineral water like distilled or reverse osmosis water and adding back measured amounts of various salts is becoming increasingly popular among home brewers. One of the likely reasons is that reverse osmosis (commonly abbreviated as RO) units have become reasonably affordable. Many grocery stores also sell RO filtered water for as low as 25 cents for the gallon. Another factor is that this process provides ultimate control over the brewing water composition. While being enthusiastically embraced by home brewers it is not as economical for larger brewers since the production of 1 liter of reverse osmosis water may require as much as 6 liters raw water though the efficiency of these units largely depends on the mineral content of the raw water and the design of the unit.

Reverse osmosis works by pushing water through a filter membrane that has openings large enough for water but not for mineral ions. The result is lower mineral water, the product water, on one side and higher mineral water, the waste water, on the other side. The process is slow and RO units are generally equipped with storage tanks that deliver water when needed. For residential "under counter" units these tanks generally hold only 1.5 - 1.7 gallons which is too little to supply the water needed in brewing. An upgrade to a 7 gallon or even larger tank is recommended. The mineral content of the water should regularly be tested with a TDS (Total Dissolved Solids) meter. These meters measure the electric resistance of the water to determine its mineral content and tend to be included when a RO unit is purchased. An increase in the RO water's mineral content means that it is time to replace the membrane. I have been using my RO unit for about 4 years now and have not had the need to replace the membrane.

RO water is not completely free of minerals. About 5 - 10% of the initial mineral content remains but the resulting water is low enough in minerals that it can be treated as being free of minerals. If so desired you may also send a sample of the RO water to a water analysis lab to get a base water profile for your brewing water additions.

When building water from scratch brewers should have the following salts on hand:

- gypsum (calcium sulfate)
- calcium chloride
- chalk (calcium carbonate)
- baking soda (sodium bicarbonate)
- Epsom salts (magnesium sulfate)

Baking soda can be found in the baking aisle and Epsom salt can be found in the health section of grocery stores. The other salts are best purchased at a home brewing store.

A digital scale for precisely measuring grams of salts is also very helpful. I have had very good success with jewelry scales (100-200g capacity, 0.01g resolution) that can be found on e-bay for ~\$20.

Water modification spreadsheet

Calculating and predicting the results of water modification can be complicated due to the chemistry involved. To simplify these calculations a water calculation spreadsheet (https://web.archive.org/web/20240805161308/http://braukaiser.com/documents/Kaiser_water_calculator.xls) has been developed. This spreadsheet supports:

- the use of simple GH&KH or detailed water analysis results

- mixing and dilution of water
- addition of salts for water treatment including dissolved chalk
- addition of slaked lime to raise mash pH
- addition of lactic acid, acidulated malt and phosphoric acid
- slaked lime treatment
- beer color based mash pH prediction

Designing and modifying brewing water

Using a water spreadsheet and determining the necessary water modifications is one of the more daunting aspects of water chemistry for brewers that want to start taking charge of their water. There are just too many variables that can be changed. Here are some guidelines that can help:

- Correct the alkalinity first. This may also require the addition of acids.
- if alkalinity needs to be raised and the sodium content of the base water is low baking soda might be used as long as the sodium level stays below 30-40 mg/l (just something I have done in the past) after that chalk is the better option.
- then make sure that the calcium level is in the desired range. Delicate beers tend to benefit from calcium levels at the low end (50 - 80 mg/l) while more flavorful beers might be better if the water has a calcium content at the upper end of the range (100 - 150 mg/l)
- after that pay attention to chloride and sulfate. Balance hop dominated beers toward sulfate and malt dominated beers toward chloride. In German beers I avoid sulfate levels higher than 90 mg/l even for the more hop dominated Pilsners.

Keep in mind that a salt always add both an anions (-) and a cations (+). You cannot add calcium without also adding chloride, sulfate for alkalinity, for example. This means some tweaking is necessary to get to the desired levels.

Various water recipes lists a few water recipes that have worked well for me in the past.

Historic and current water profiles of major brewing cities may also be used as a guideline for a particular beer. But it is important to understand that these waters may have undergone water treatment before being used to brew a style that originated from that region. If the historic water profile fits the beer that you are brewing, i.e. the mash pH is in an acceptable range, you may want to target this water since brewers may most likely have been using this water for their beers.

A classic example for this is the water profile for Munich. While this water is great for brewing a Munich Dunkel or a Doppelbock it is the wrong water for brewing a Munich Helles. This largely Pilsner malt based beer requires water with low residual alkalinity. To get to the desired mash pH Munich brewers likely precipitate alkalinity from the water using lime and in addition to that use sour fermented wort (*Sauergut*) to correct the mash pH even lower. The Paulaner brewery, for example, doesn't even have to decarbonate their water. They are getting very soft (~ 2 dH, 40 ppm as CaCO₃) brewing water from a deep well on their property.

Salt and acid additions to the mash



Mash pH and later wort pH can also be adjusted on the fly. This is generally done by adding either salts or acids to bring the pH back into the desired range after it has been tested and was found to be too high or too low.

Since the malt is the dominating pH buffer the calculation of the necessary acid or salts are best based on the grist weight while mash thickness (i.e. the amount of mash water used) matters little.

All salts and acids that can be added to the water can also be added to the mash. If salts are added only to the mash their effective concentration in the beer will be lower since the additional water used for sparging will dilute the added minerals in the kettle.

While it is practical to treat water and grist such that they achieve the desired mash pH at dough-in, many brewers find it easier to measure the pH of a mash sample 5-10 min after dough-in and base the addition of salts and acids based on how far away this pH is from the targeted pH. This works particularly well for brewers who rarely repeat recipes and thus have less past data that could be used to estimate the necessary water and mash treatments.

While there doesn't appear to be any difference between treating the water and treating the mash with respect to the pH the mash will settle at, there can be a slight difference in expression of enzymatic activity. In particular in single infusion mashing where the bulk of the fermentable sugars is created by beta-amylase early during the mash before large amounts of that enzyme fall victim to the mash temperature. This enzyme has a pH optimum between 5.3 - 5.6 ^[7] and if the pH of the mash is sufficiently far off this optimum during the first 10-20 min the amount of fermentable sugars produced and with it the fermentability of the created wort may suffer.

Table 3 contains a summary of the possible pH treatments that can be made to the mash or to the strike water.

	for 1 kg		for 1 lb		notes
lactic acid (88% w/w)	0.25 - 0.42	ml	0.125 - 0.21	ml	While the preferred acid for pH treatment in German beers, the addition should not exceed 1 ml/kg (0.5 ml/lb) in order prevent a noticeable flavor impact. This is based on the "no more than 4% acid malt rule"
acidulated malt (Sauermalz) (3% w/w lactic acid)	9 - 15	g	4.5 - 7.5	g	the acidulated malt content should not exceed 4 % of the grist weight
37% hydrochloric acid	0.24 - 0.32	ml	0.12 - 0.16	ml	Hydrochloric acid is a very aggressive and corrosive acid and needs to be handled with care
phosphoric acid (10% w/w)	2.94 - 3.92	ml	1.47 - 1.96	ml	Phosphoric acid is a natural choice for mash acidification since the malt itself already adds large amounts of phosphates to the mash.
phosphoric acid (85% w/w)	0.2 - 0.27	ml	0.1 - 0.135	ml	Phosphoric acid may be available in different strength, hence the suggestion for 10% and 85% w/w concentrations
gypsum (calcium sulfate) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	1 - 1.3	g	0.5 - 0.65	g	The amount of gypsum given here is equivalent to about 30-40 mg/l additional calcium and 74-96 mg/l additional sulfate in the brewing water *
calcium chloride $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$	0.9 - 1.1	g	0.45 - 0.55	g	The amount of calcium chloride given here is equivalent to about 32-39 mg/l additional calcium and 57-70 mg/l additional chloride in the brewing water *

mash additions to raise mash pH by 0.1

	for 1 kg		for 1 lb		notes
chalk, calcium carbonate (CaCO_3)	0.83 - 1.6	g	0.415 - 0.8	g	Chalk does not completely dissolve during mashing which limits its effectiveness and predictability for mash pH adjustment, hence the wide spread. It is possible that the pH change caused by this addition is far less than 0.1. The amount given here is equivalent to an additional 44 - 85 mg/l calcium in the brewing water *
baking soda, sodium bicarbonate (NaHCO_3)	0.42 - 0.67	g	0.21 - 0.335	g	baking soda works more predictable than chalk with respect to changing the mash pH. It does however adds sodium ions which are less desirable than calcium ions. The amount given here is equivalent to about 15 - 24 mg/l additional sodium in the brewing water *
slaked lime, calcium hydroxide (Ca(OH)_2)	0.21 - 0.25	g	0.105 - 0.125	g	slaked lime is more soluble than chalk and produces a more predictable change in mash pH. It also adds calcium to the mash and is therefore an alternative to the use of chalk for increasing mash pH. The amount given here is equivalent to about 15 - 17 mg/l additional calcium in the brewing water *. Slaked lime is caustic and needs to be handled with care
caustic soda, sodium hydroxide (NaOH)	0.16 - 0.2	g	0.08 - 0.1	g	the use of sodium hydroxide does not provide any advantages over the use of baking soda. It is listed here only for completeness. Sodium hydroxide is caustic and needs to be handled with care.

* The effective water ion increase was estimated for a 12 P beer brewed with 85 % efficiency into the kettle and 15 % boil-off

Table 3 - Guidelines for adding acids and salt to the mash to lower or raise the mash pH by 0.1 pH units. These guidelines have been developed from small scale mash experiments and observations during brewing

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